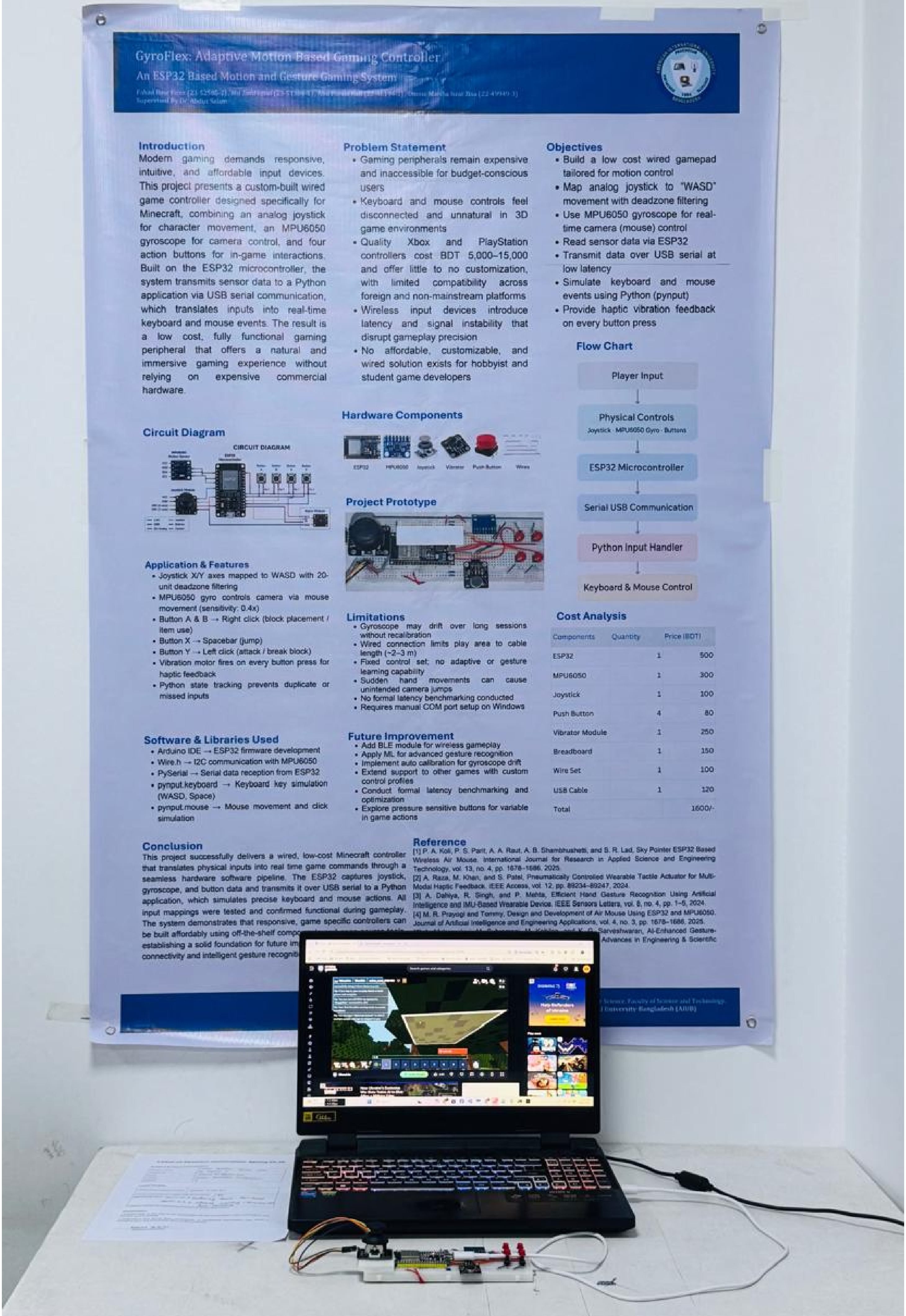

GyroFlex: Adaptive Motion Based Gaming Controller

An ESP32 Based Motion and Gesture Gaming System

[illegible]
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Introduction

Modern gaming demands responsive, intuitive, and affordable input devices. This project presents a custom-built wired game controller designed specifically for Minecraft, combining an analog joystick for character movement, an MPU6050 gyroscope for camera control, and four action buttons for in-game interactions. Built on the ESP32 microcontroller, the system transmits sensor data to a Python application via USB serial communication, which translates inputs into real-time keyboard and mouse events. The result is a low cost, fully functional gaming peripheral that offers a natural and immersive gaming experience without relying on expensive commercial hardware.

Circuit Diagram

Application & Features

- Joystick X/Y axes mapped to WASD with 20-unit deadzone filtering
- MPU6050 gyro controls camera via mouse movement (sensitivity: 0.4x)
- Button A & B → Right click (block placement / item use)
- Button X → Spacebar (jump)
- Button Y → Left click (attack / break block)
- Vibration motor fires on every button press for haptic feedback
- Python state tracking prevents duplicate or missed inputs

Software & Libraries Used

- Arduino IDE → ESP32 firmware development
- Wire.h → I2C communication with MPU6050
- PySerial → Serial data reception from ESP32
- pynput.keyboard → Keyboard key simulation (WASD, Space)
- pynput.mouse → Mouse movement and click simulation

Problem Statement

- Gaming peripherals remain expensive and inaccessible for budget-conscious users
- Keyboard and mouse controls feel disconnected and unnatural in 3D game environments
- Quality Xbox and PlayStation controllers cost BDT 5,000–15,000 and offer little to no customization, with limited compatibility across foreign and non-mainstream platforms
- Wireless input devices introduce latency and signal instability that disrupt gameplay precision
- No affordable, customizable, and wired solution exists for hobbyist and student game developers

Hardware Components

Project Prototype

Limitations

- Gyroscope may drift over long sessions without recalibration
- Wired connection limits play area to cable length (~2–3 m)
- Fixed control set; no adaptive or gesture learning capability
- Sudden hand movements can cause unintended camera jumps
- No formal latency benchmarking conducted
- Requires manual COM port setup on Windows

Future Improvement

- Add BLE module for wireless gameplay
- Apply ML for advanced gesture recognition
- Implement auto calibration for gyroscope drift
- Extend support to other games with custom control profiles
- Conduct formal latency benchmarking and optimization
- Explore pressure sensitive buttons for variable in game actions

Objectives

- Build a low cost wired gamepad tailored for motion control
- Map analog joystick to "WASD" movement with deadzone filtering
- Use MPU6050 gyroscope for real-time camera (mouse) control
- Read sensor data via ESP32
- Transmit data over USB serial at low latency
- Simulate keyboard and mouse events using Python (pynput)
- Provide haptic vibration feedback on every button press

Flow Chart

Player Input
↓
Physical Controls
Joystick · MPU6050 Gyro · Buttons
↓
ESP32 Microcontroller
↓
Serial USB Communication
↓
Python Input Handler
↓
Keyboard & Mouse Control

Cost Analysis

Components	Quantity	Price (BDT)
ESP32	1	500
MPU6050	1	300
Joystick	1	100
Push Button	4	80
Vibrator Module	1	250
Breadboard	1	150
Wire Set	1	100
USB Cable	1	120
Total		1600/-

Conclusion

This project successfully delivers a wired, low-cost Minecraft controller that translates physical inputs into real time game commands through a seamless hardware software pipeline. The ESP32 captures joystick, gyroscope, and button data and transmits it over USB serial to a Python application, which simulates precise keyboard and mouse actions. All input mappings were tested and confirmed functional during gameplay. The system demonstrates that responsive, game specific controllers can be built affordably using off-the-shelf comp
establishing a solid foundation for future im
connectivity and intelligent gesture recogni

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